2024春线性代数期中（回忆版）

1. Multiple Choice. Only one choice is correct.

(1) Let $\alpha_1=\begin{bmatrix}2\\3\\1\end{bmatrix}$, $\alpha_2=\begin{bmatrix}1\\-1\\2\end{bmatrix}$, $\alpha_3=\begin{bmatrix}7\\3\\c\end{bmatrix}$.

If $\alpha_1,\alpha_2,\alpha_3$ are linearly dependent, then c equals.

(A) 5 (B) 6 (C) 7 (D) 8

(2) Let A be an $m\times n$ real matrix and b be an $m\times 1$ real column vector. Which of the following statements is correct?

(A) If $Ax=b$ does not have any solution, then $Ax=0$ has only the zero solution.

(B) If $Ax=0$ has infinitely many solutions, then $Ax=b$ has infinitely many solutions.

(C) If $m<n$, both $Ax=b$ and $Ax=0$ have infinitely many solutions.

(D) If the rank of A is n, then $Ax=0$ has only the zero solution.

(3) For which value of k does the system

$$\begin{cases}x_1+2x_2-4x_3+3x_4=0\\x_1+3x_2-2x_3-2x_4=0\\x_1+5x_2+(5-k)x_3-12x_4=0\end{cases}$$

have exactly two free variables?

(A) 5 (B) 4 (C) 3 (D) 2

(4) Let $u,v\in R^3$ and $\lambda\in R$. Which of the following statement is false?

(A) If u and v are nonzero vectors satisfying $u^Tv=0$, then u and v are linearly independent.

(B) If $u+v$ is orthogonal to $u-v$, then $\|u\|=\|v\|$.

(C) $u^Tv=0$ if and only if $u=0$ or $v=0$.

(D) $\lambda v=0$ if and only if $v=0$ or $\lambda=0$.

(5) Let A and B be two $n\times n$ matrices. Which of the following assertions is false?

(A) If A,B are symmetric matrices, then AB is a symmetric matrix.

(B) If A,B are invertible matrices, then AB is an invertible matrix.

(C) If A,B are permutation matrices, then AB is a permutation matrix.

(D) If A,B are upper triangular matrices, then AB is an upper triangular matrix.

2. Fill in the blanks.

(1) Let $A=\begin{bmatrix}1&0&0\\a&1&0\\b&3&2\end{bmatrix}$, $a,b\in R$.

Then $A^{-1}=$ _____.

(2) Let A be a 4×3 real matrix with rank 2 and $B=\begin{bmatrix}1&0&2\\0&2&0\\-1&0&3\end{bmatrix}$. Then the rank AB is _____.

(3) Let $A=\begin{bmatrix}1&-1&1\\-1&1&-1\\2&-2&2\end{bmatrix}$. Then $A^{2024}=$ _____.

(4) Consider the system of linear equations:

$$Ax=b:\begin{cases}x=2\\y=3\\x+y=6\end{cases}$$

The least-squares solution for the system is _____.

3. Let $A=\begin{bmatrix}1&-2&3\\2&-5&1\\1&-4&-7\end{bmatrix}$

Find an LU factorization of A.

4. Consider the following 4×5 matrix A and 4-dimensional column vector b:

$$A=\begin{bmatrix}0&2&4&1&6\\0&1&1&1&3\\0&4&10&1&2\\0&-1&-5&1&7\end{bmatrix},\ b=\begin{bmatrix}3\\2\\-5\\10\end{bmatrix}$$

(a) Find a basis for each of the four fundamental subspace of A.

(b) Find the compelete solution to $Ax=b$.

5. Let $A=\begin{bmatrix}1&1\\0&2\end{bmatrix}$ and T be the linear transformation from $R^{2\times2}$ to $R^{2\times2}$ defined by $T(X)=XA+AX,\ X\in R^{2\times2}$

where $R^{2\times2}$ denotes the vector space consisting of all 2×2 real matrices.

(a) Find the matrix representation of T with respect to the following ordered basis

$$v_1=\begin{bmatrix}1&0\\0&0\end{bmatrix},\ v_2=\begin{bmatrix}0&1\\0&0\end{bmatrix},\ v_3=\begin{bmatrix}0&0\\1&0\end{bmatrix},\ v_4=\begin{bmatrix}0&0\\0&1\end{bmatrix}.$$

(b) Find a matrix B such that

$$T(B)=\begin{bmatrix}0&0\\0&0\end{bmatrix}$$

(c) Find a matrix C such that

$$T(C)=\begin{bmatrix}1&2\\3&4\end{bmatrix}$$

6. Let A,B be two $n\times n$ real matrices satisfying $A^2=A$ and $B^2=B$. Show that if $(A+B)^2=A+B$, then $AB=0$. When 0 denotes the $n\times n$ zero matrix.

7. Let A be a 3×2 matrix, B be a 2×3 matrix such that

$$AB=\begin{bmatrix}8&0&-4\\-\frac{3}{2}&9&-6\\-2&0&1\end{bmatrix}$$

(a) Compute $(AB)^2$.

(b) Find BA.